Princess Sumaya University for Technology - King Hussein School for Computing Sciences

CS11313 - **Algorithms Design and Analysis** - **Spring** 2023

Quiz # 1

Name:	**ID:**

Q1. Assume the running time of some algorithm satisfies the following relations: $T(n) = \Omega(n^2 \log(n))$ and $T(n) = o(n^3 \log(n))$, which of the following functions can $T(n)$ be? Circle all that apply.

n^2 $\quad$ $n^{2.1}$ $\quad$ n^3 $\quad$ $n^{3.1}$ $\quad$ $n \log(n!)$ $\quad$ $n^2 \log(n!)$

Q2. Fill in the following table. Each box should have a YES or NO indicating whether or not the condition at the top of the row is true.

$f(n)$	$g(n)$	$f = O(g)$	$f = \Omega(g)$	$f = \Theta(g)$	$f = o(g)$	$f = \omega(g)$
$n^{\frac{1}{10}}$	$\log_{10}(n)$	**NO**	**YES**	**NO**	**NO**	**YES**
$n^{\log_2(10)}$	$10^{\log_2(n)}$	**YES**	**YES**	**YES**	**NO**	**NO**
10	$\log_2(10)$	**YES**	**YES**	**YES**	**NO**	**NO**
$\sum_{i=0}^{\log_3 n} 3^i$	$\sum_{i=0}^{n} 2^i$	**YES**	**NO**	**NO**	**YES**	**NO**
$\log_2(n\sqrt{n})$	$\log_2(n \log_2 n)$	**YES**	**YES**	**YES**	**NO**	**NO**

Q3. Show that $kn = O(\frac{n}{k})$, where k is a constant greater than 2.

We need to show that there exist two constants c and n0, such that kn <= cn/k for all n >= n0.

We can pick c = k^2:
kn <= (k^2)n/k <= kn

This is clearly true for all n >= 1